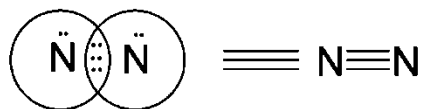
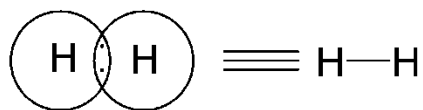


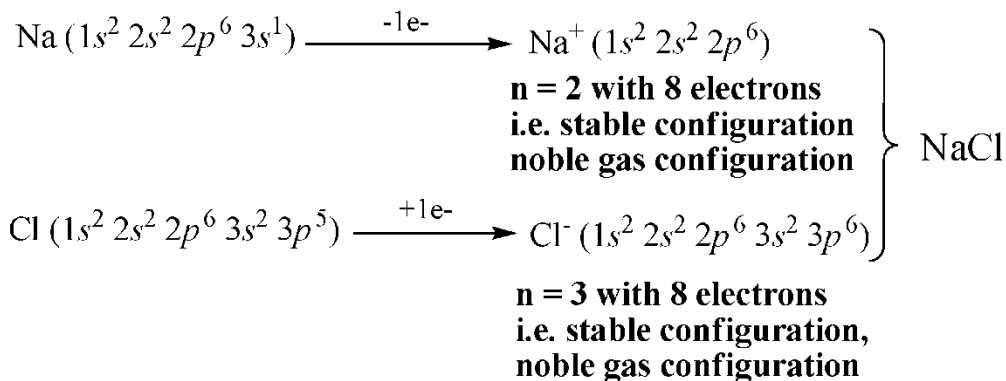
Hybridization and MO Theory

Types of bonds

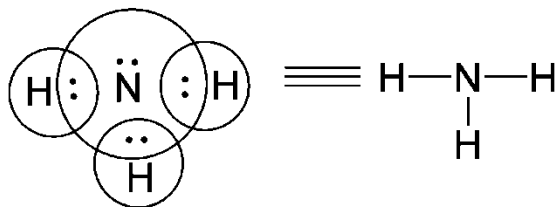
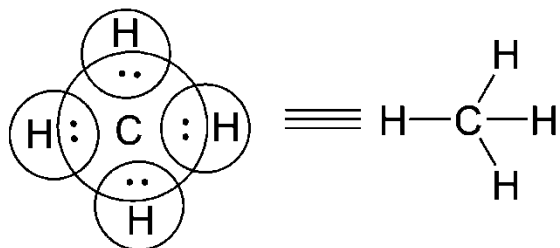
Covalent Homoatomic bonding



Ionic bonding



Partial Covalent Heteroatomic bonding



Fajan's Rule:

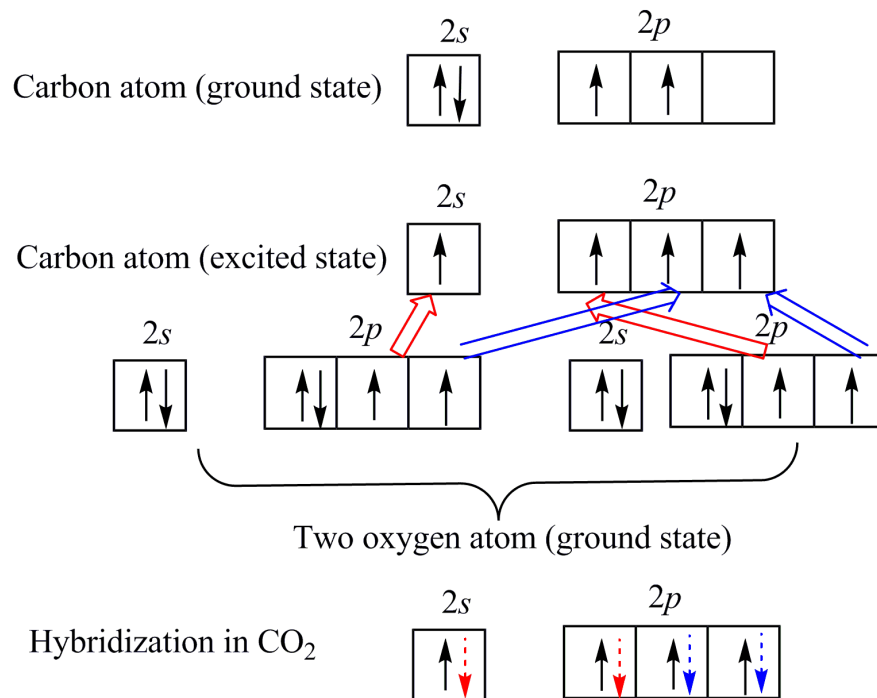
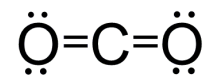
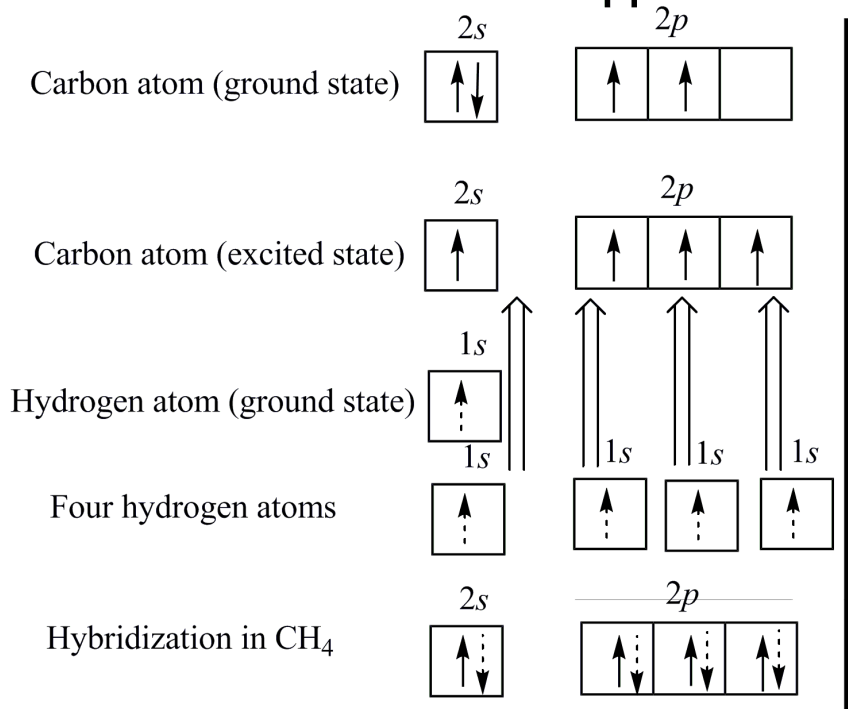
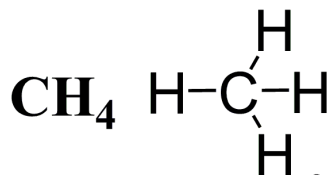
An ion will be formed most easily when;

- (i) the electronic structure of the ion is stable,
- (ii) the charge on the ion is small, and
- (iii) a small atom forms the anion and a large atom forms the cation.

Increased covalence is favored by;

- (i) high charge on the ions,
- (ii) small size of the cation and large size of the anion,
- (iii) cations with 18 electron structures, i.e. noble-gas configuration. Noble gas configuration of the cation produces better shielding and less polarizing power.

Hybridization



Red color arrows forms σ -bonds
Blue color arrows forms π -bonds

Draw structures of CH₄, CO₂ and H₂O
Different s character and p character so angle varies
With different hybridization

Find out for H₂O

Molecular Orbital Theory

$\psi(\text{MO}_b) = \psi(A) + \psi(B)$; $\psi(\text{MO}_b)$ = bonding MO wave function

and

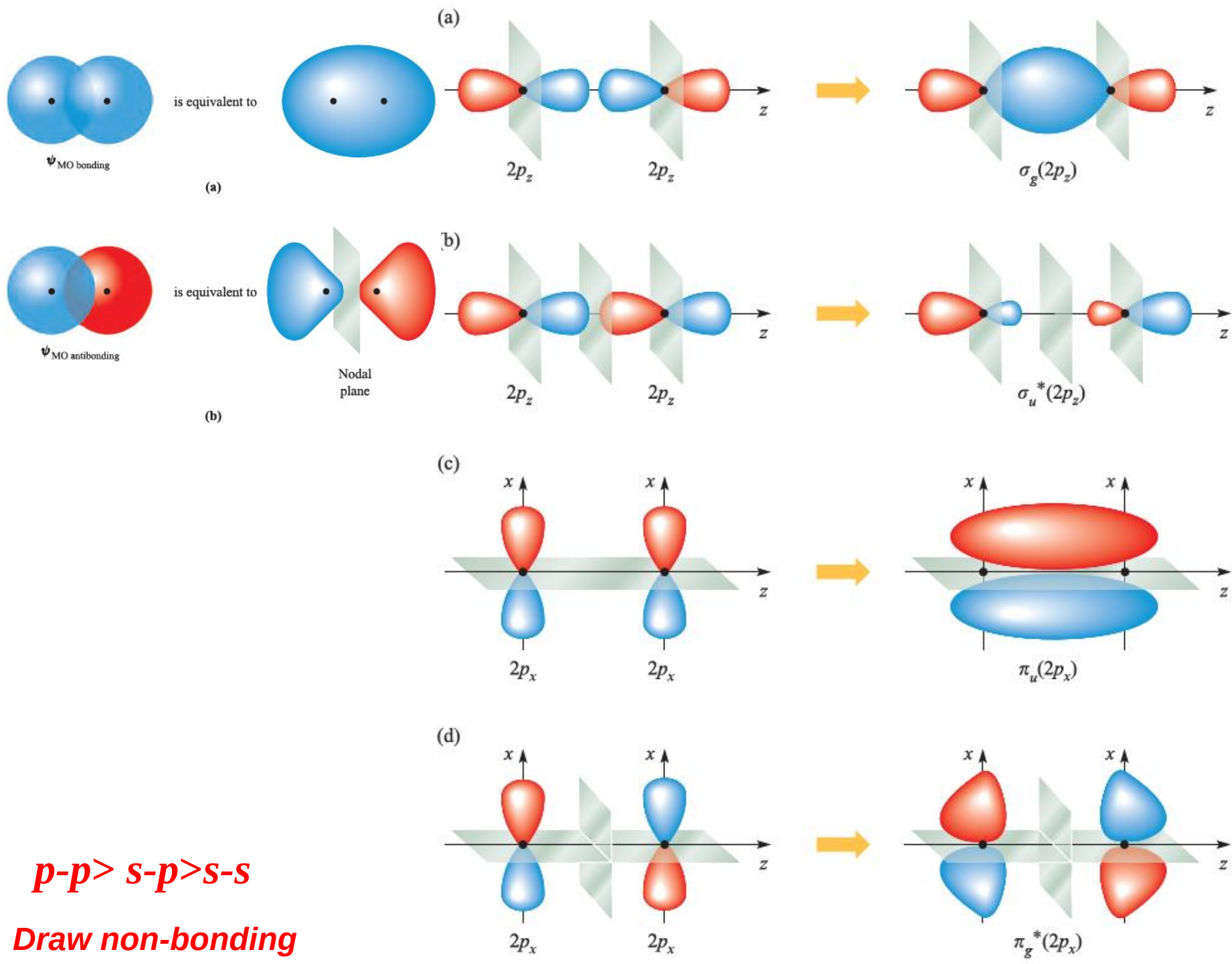
$\psi(\text{MO}_{ab}) = \psi(A) - \psi(B)$; $\psi(\text{MO}_{ab})$ = anti-bonding MO wave function

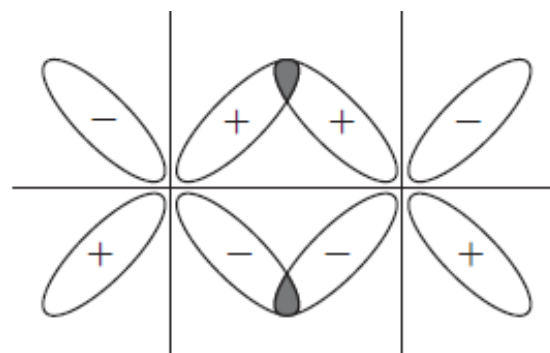
Conditions for the favorable combination of atomic orbitals:

(A) Atomic orbitals must have comparable energies.

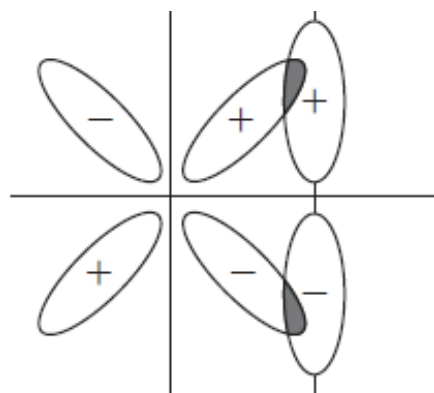
(B) They must overlap appreciably.

(C) They must have same symmetry with respect to the bonding molecular axis, *i.e.* if the atomic orbitals are perpendicular to each other there will be no overlap or linear combination between them.

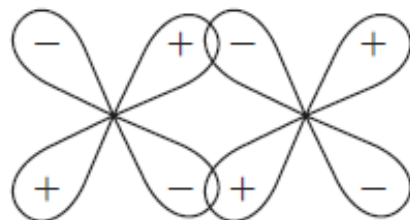
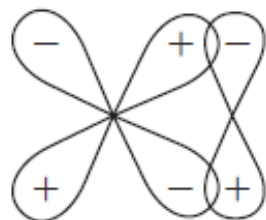
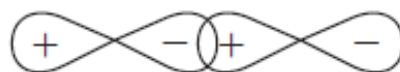
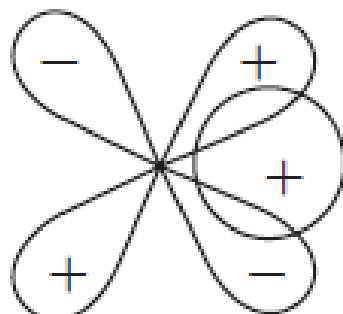
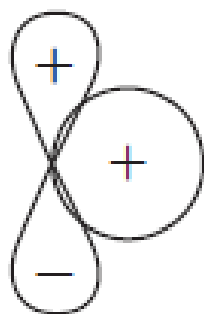




d-d

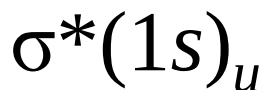
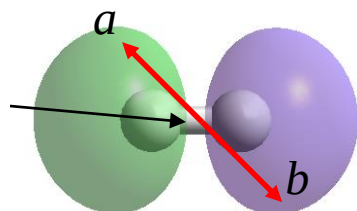


d-p

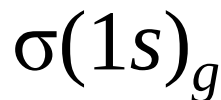
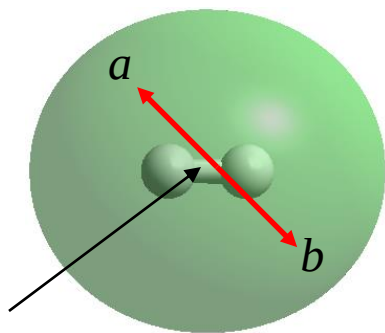


g or *u* symmetry of molecular orbitals:

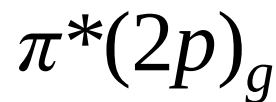
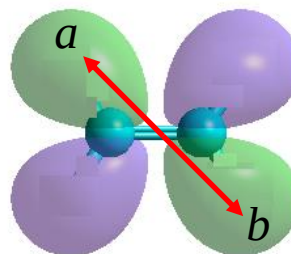
not a center
of inversion
 $a \neq b$
(sign of wave-
function is
opposite)



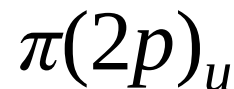
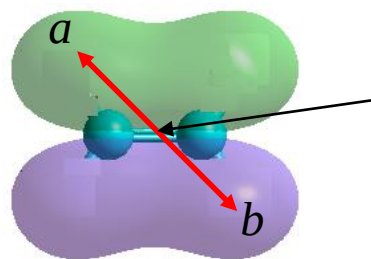
center of
inversion
 $a = b$



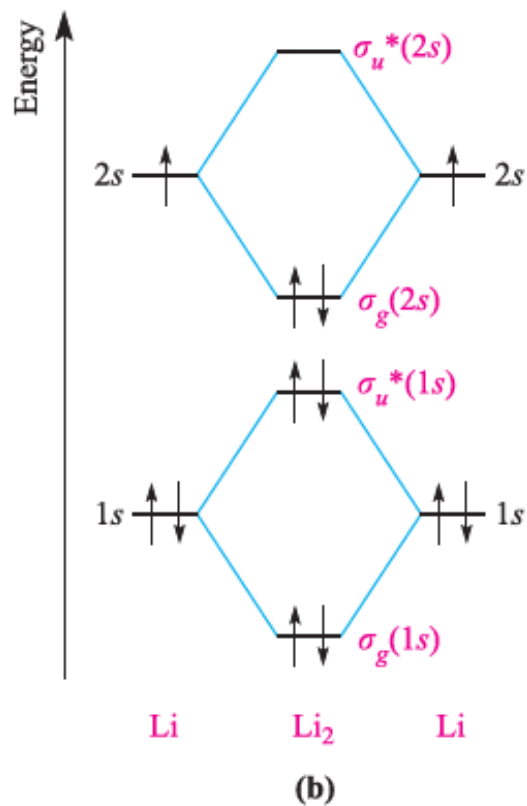
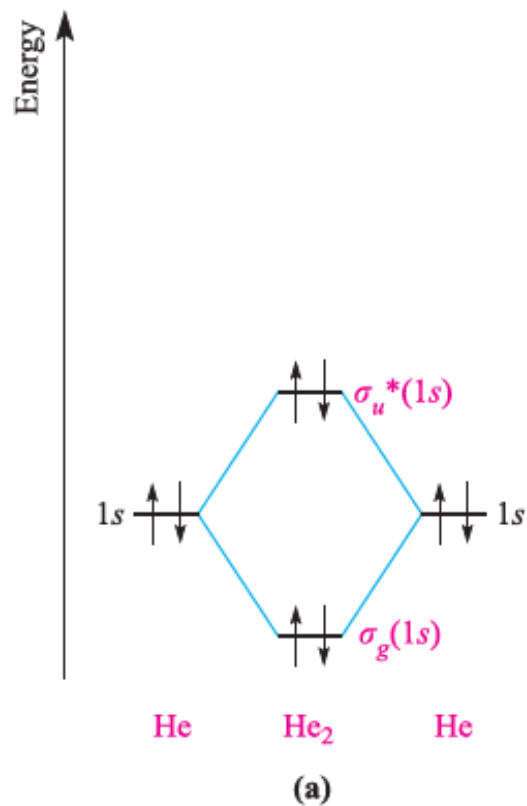
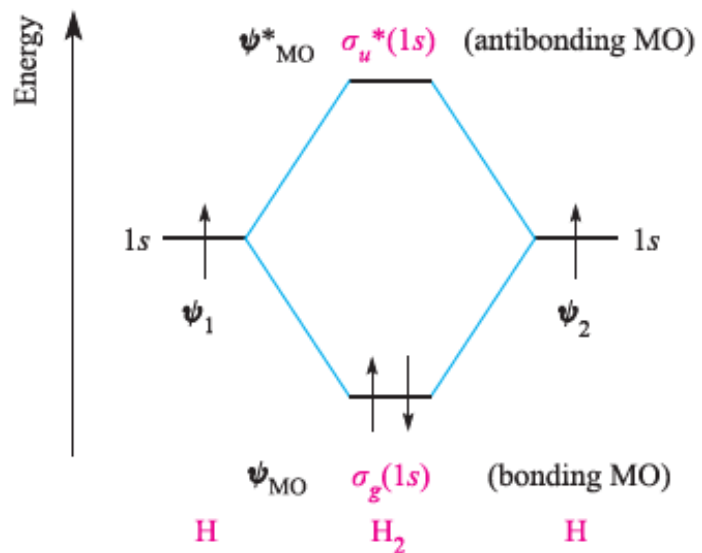
center of
Inversion
 $a = b$

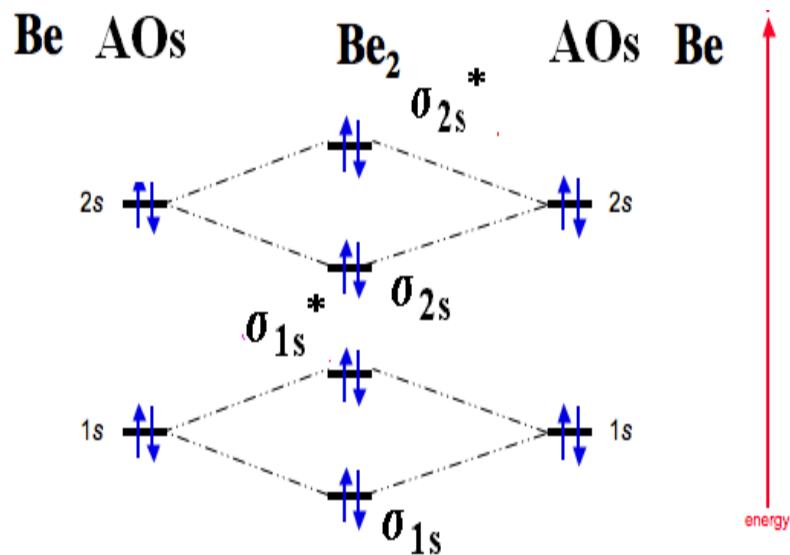


not a center of
inversion
 $a \neq b$

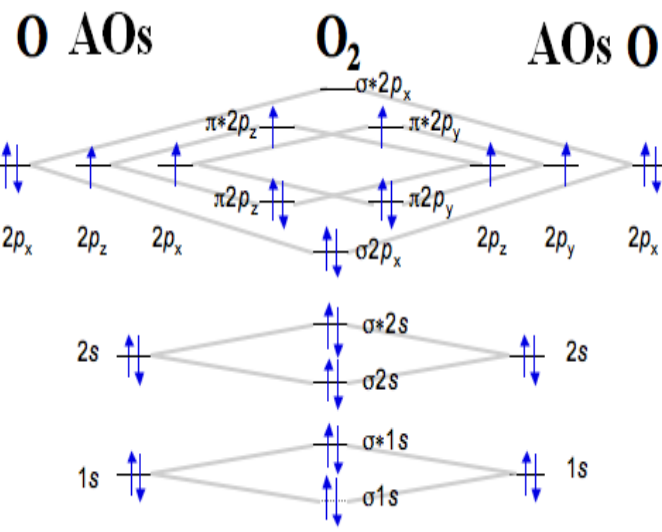
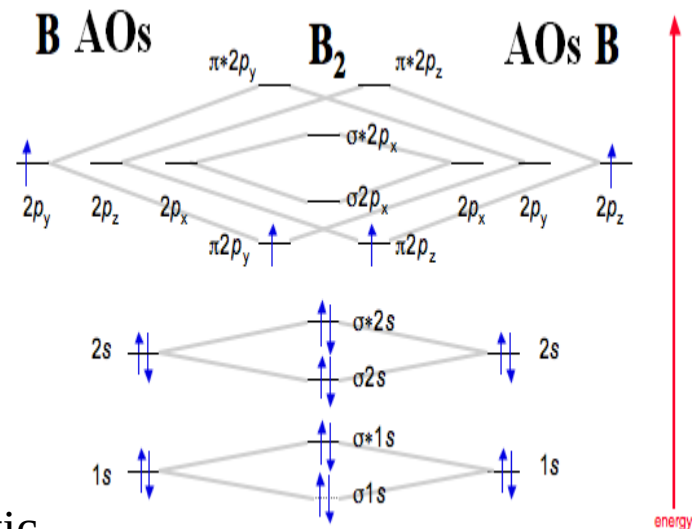


The test for whether an MO is *g* or *u* is to find the possible center of inversion of the MO. If two lines drawn out at 180° to each other from the center, and of equal distances, strike identical points (*a* and *b*), then the orbital is *g*.





Paramagnetic



Paramagnetic

Variation of the energies of the 2s and 2p orbitals

